

e.g

now we look for the global max and min of
 $f(x, y, z) = 2x + y - z$ over the closed unit ball
 $\bar{X} = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$.

1) $\text{bd}(\bar{X}) = \{(x, y, z) : x^2 + y^2 + z^2 = 1\}$

2) Critical points of f inside $\bar{X}$:

$\vec{\nabla} f = (2, 1, -1)$ is no critical points.

3) Evaluate f at critical points and boundary:

We only care about the boundary, and
what we want is to optimize $f(x, y, z) = 2x + y - z$
subject to $g(x, y, z) = x^2 + y^2 + z^2 = 1$

Let's use Lagrange multipliers:

We want points (x, y, z) where

$$\vec{\nabla} f = (2, 1, -1) = \lambda \vec{\nabla} g = \lambda (2x, 2y, 2z)$$

Get eqns. $2 = 2x\lambda$ $1 = 2y\lambda$ $-1 = 2z\lambda$

$$2x\lambda = 2 = 4y\lambda = -4z\lambda$$

either $\lambda = 0$ (which would mean $\lambda \vec{\nabla} g = 0 \neq \vec{\nabla} f$),

or $2x = 4y = -4z$

so $x = 2y$ and $y = -z$

the constrained critical point must look

like $(2y, y, -y)$ for some $y \in \mathbb{R}$

and must satisfy the constraint:

$$1 = x^2 + y^2 + z^2 = 4y^2 + y^2 + y^2 = 6y^2$$

$$y = \pm \sqrt{1/6}$$

solution $(2/\sqrt{6}, 1/\sqrt{6}, -1/\sqrt{6})$ or $(-2/\sqrt{6}, -1/\sqrt{6}, 1/\sqrt{6})$

Evaluate: $f(2/\sqrt{6}, 1/\sqrt{6}, -1/\sqrt{6}) = \sqrt{6}$ $f(-2/\sqrt{6}, -1/\sqrt{6}, 1/\sqrt{6}) = -\sqrt{6}$

4) global max is $\sqrt{6}$ global min is $-\sqrt{6}$

e.g. Q.6 Find the point on the sphere $x^2 + y^2 + z^2 = 16$ that is closest (and the point that is farthest) from $(3, 1, -1)$

$$f(x, y, z) = (x-3)^2 + (y-1)^2 + (z+1)^2$$

constraint $g(x, y, z) = x^2 + y^2 + z^2 \quad g(\vec{a}) = 16$

We can use Lagrange multipliers

We find points where $\vec{\nabla} g|_{\vec{a}} = 0$ or $\vec{\nabla} f|_{\vec{a}} = \lambda \vec{\nabla} g|_{\vec{a}}$

$$\vec{\nabla} f = (2(x-3), 2(y-1), 2(z+1))$$

$$\vec{\nabla} g = (2x, 2y, 2z)$$

$\vec{\nabla} g|_{\vec{a}} = \vec{0}$ only at $\vec{a} = \vec{0}$
but $\vec{0}$ is not on the sphere
so we disregard $\vec{a} = \vec{0}$

so suppose $\vec{\nabla} f = \lambda \vec{\nabla} g$

$$\begin{aligned} 2(x-3) &= \lambda 2x, & 2(y-1) &= \lambda 2y, & 2(z+1) &= \lambda 2z \\ 2x-6 &= 2\lambda x, & 2y-2 &= 2\lambda y, & 2z+2 &= 2\lambda z \\ x(1-\lambda) &= 3, & y(1-\lambda) &= 1, & z(1-\lambda) &= -1 \end{aligned}$$

from here $x = 3y, \quad z = -y$

so the point we are looking for is of the form

$(3y, y, -y)$ and must satisfy $g(3y, y, -y) = 16$
 $\rightarrow 9y^2 + y^2 + y^2 = 16 \quad 11y^2 = 16$
 $y^2 = \frac{16}{11} \quad y = \pm \frac{4}{\sqrt{11}}$

constrained critical points are $\pm \left(\frac{12}{\sqrt{11}}, \frac{4}{\sqrt{11}}, -\frac{4}{\sqrt{11}} \right)$

$$\begin{aligned} f\left(\frac{12}{\sqrt{11}}, \frac{4}{\sqrt{11}}, -\frac{4}{\sqrt{11}}\right) &= \left(\frac{12}{\sqrt{11}} - 3\right)^2 + \left(\frac{4}{\sqrt{11}} - 1\right)^2 + \left(-\frac{4}{\sqrt{11}} + 1\right)^2 \\ &= \frac{(12 - 3\sqrt{11})^2}{11} + \frac{(4 - \sqrt{11})^2}{11} + \frac{(-4 + \sqrt{11})^2}{11} \end{aligned}$$

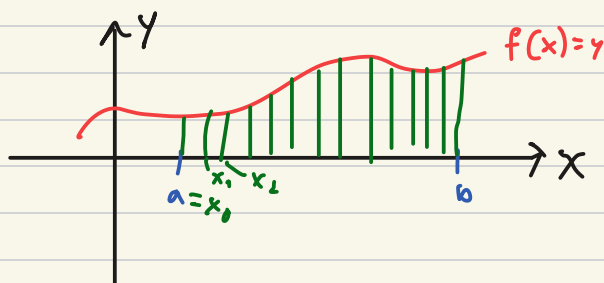
Integrals

From Calc I: Fundamental Theorem of Calculus:

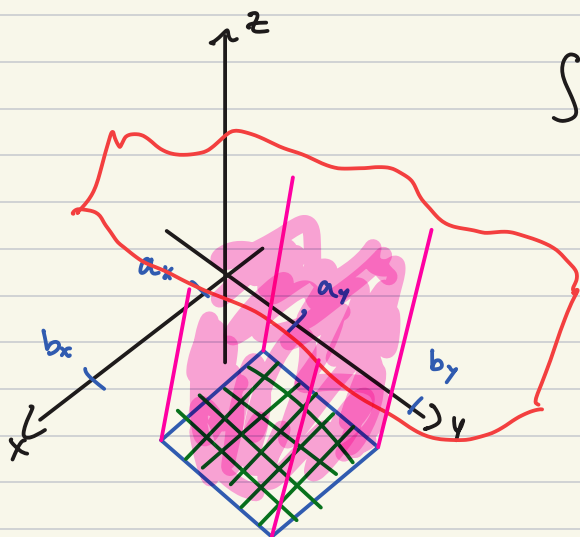
If $f: [a, b] \rightarrow \mathbb{R}$ is continuous then:

1) The function $F: [a, b] \rightarrow \mathbb{R}$ defined by $F(x) = \int_a^x f(t) dt$ is differentiable and $F'(x) = f(x)$ for all $x \in [a, b]$
(we say F is an antiderivative of f)

2) For any function F that is an antiderivative of f (i.e. $F'(x) = f(x)$) we have $\int_a^b f(x) dx = F(x) \Big|_{x=a}^{x=b} = F(b) - F(a)$

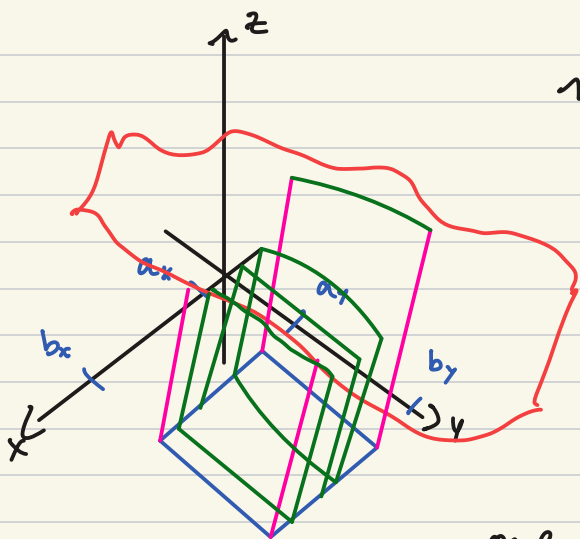


$$\int_a^b f dx = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(x_i) \Delta x_i$$



$$\int_{a_x}^{b_x} \left(\int_{a_y}^{b_y} f(x, y) dy \right) dx$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \sum_{i,j=1}^n f(x_i) \Delta y_i \Delta x_i$$



or slices
 slabs: we cut the volume
 by sections $x = c$

each slab has
 area $\int_{a_y}^{b_y} f(x_i, y) dy$
 and depth Δx_i

over a rectangular region

$$\{(x, y) : a_x \leq x \leq b_x, a_y \leq y \leq b_y\}$$

we can cut slabs perpendicular to x , or
 perpendicular to y and
 the integral (limit of Riemann sum) is
 the same

e.g. $f(x, y) = 2x + y$ we want to find

the volume under this surface over
 the rectangle $\{(x, y) : 2 \leq x \leq 4, 3 \leq y \leq 6\}$

write this as a double integral:

1) sections perpendicular to x

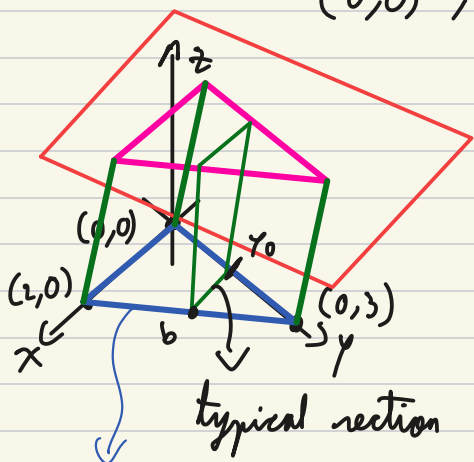
$$V = \int_2^4 \int_3^6 f(x, y) dy dx$$

2) sections perpendicular to y

$$\int_3^6 \int_2^4 f(x, y) dx dy$$

e.g. Volume of the region under $f(x,y)=1$

over the triangular region with vertices
 $(0,0)$, $(2,0)$, $(0,3)$



take slices perpendicular to y

typical section: x runs over $[0, b]$

this line has eqn. $y = 3 - \frac{3}{2}x$ or $b = 2 - \frac{2}{3}y$
 $x = 2 - \frac{2}{3}y$

$$\int_0^3 \int_0^{2-\frac{2}{3}y} 1 \, dx \, dy = \int_0^2 \int_0^{3-\frac{3}{2}x} 1 \, dy \, dx$$